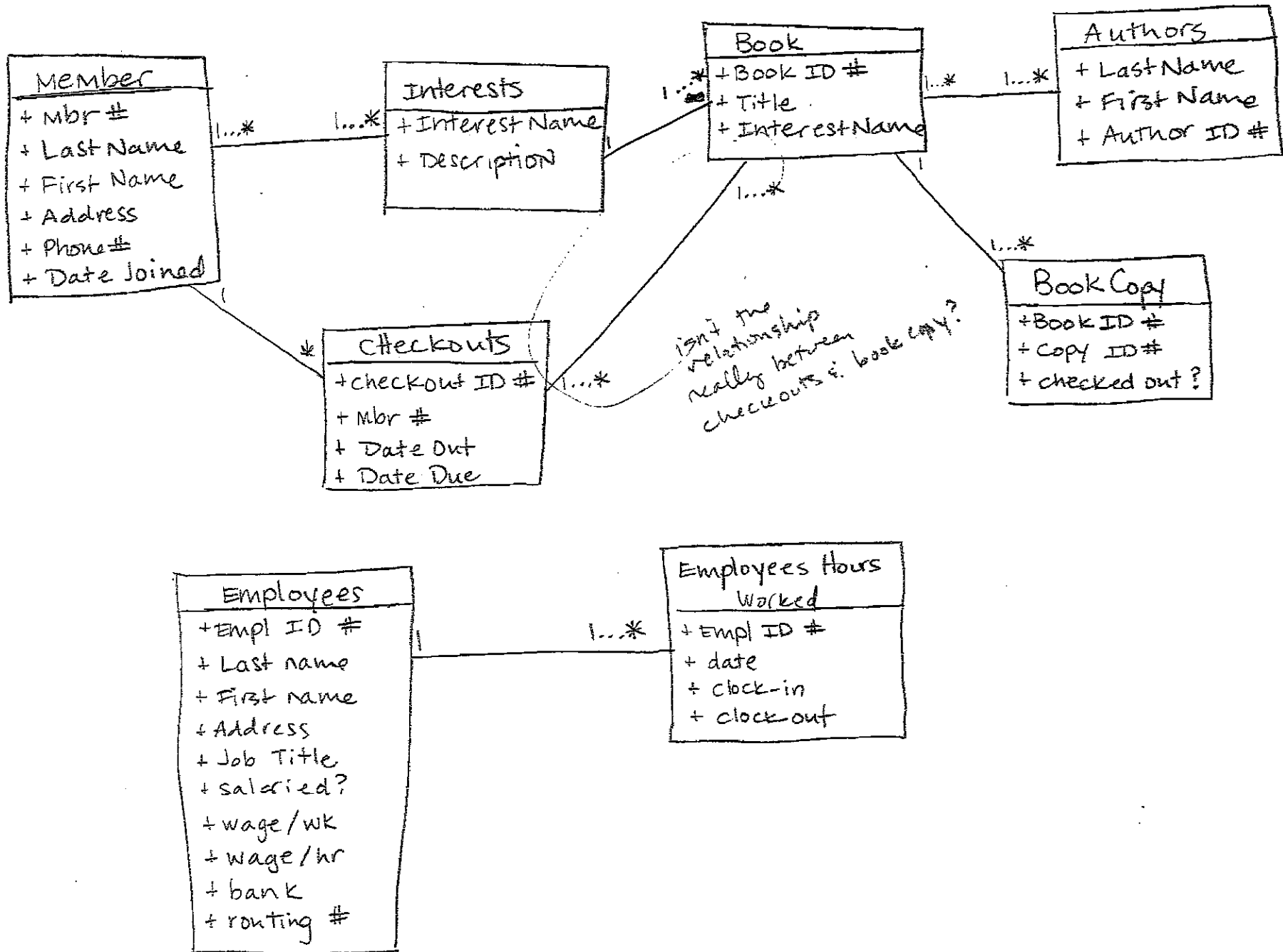


step 1

Library Database.
create E-R diagram and
ensure it's in 3NF.

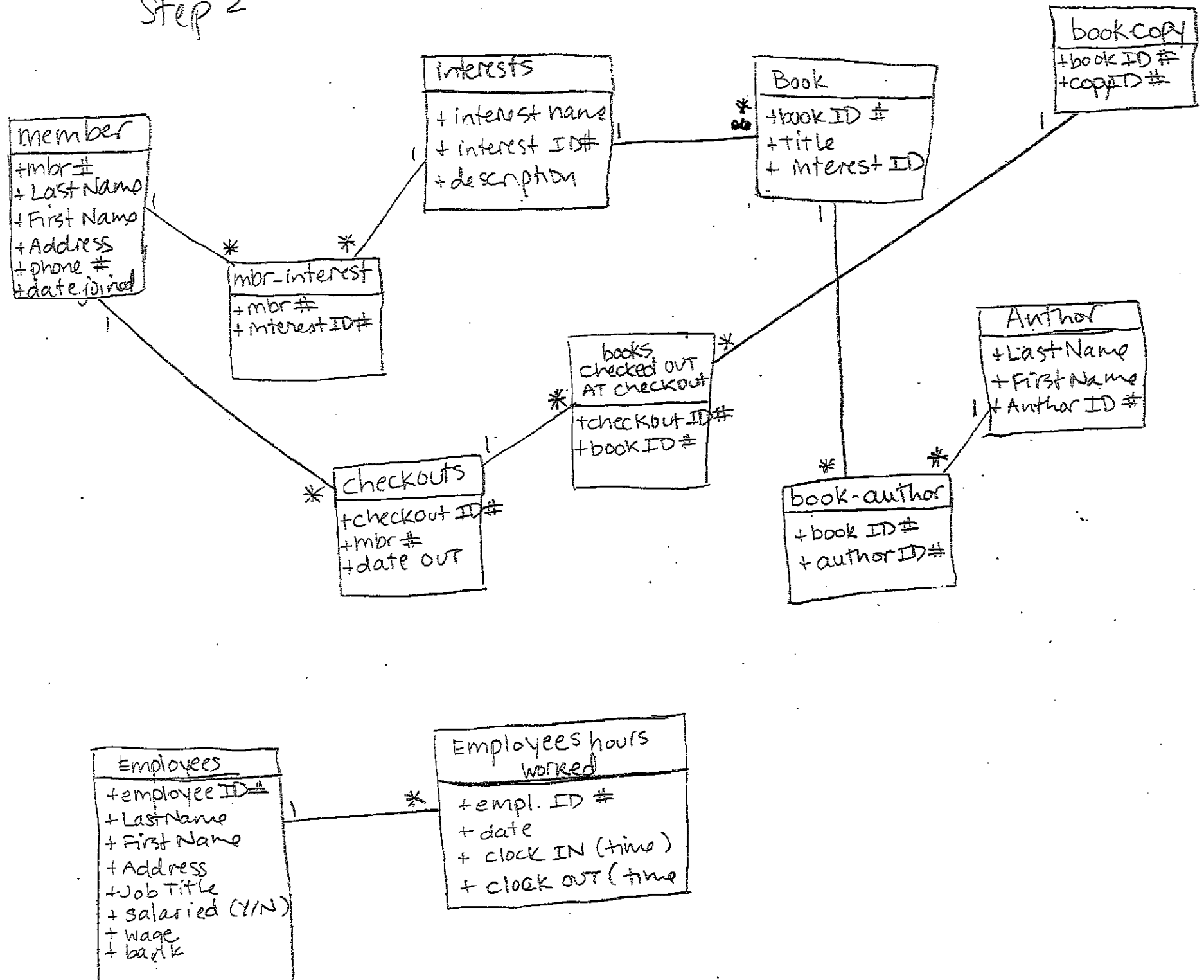


Assumptions I made:

- I included everything from the problem statement in the book.
- I assumed that the library has (or can have) multiple copies of each book.

For All of the many-to-many relationships, we have to create a third table. This 3rd table must include at least the primary keys from each of the original tables. There can be additional fields as desired/needed.

Step 2



Step 3.

MEMBER Table:

mbr #	LastName	FirstName	Address	PhoneNum	DateJoined
*					

INTEREST Table:

InterstName	Description
*	

*PrimaryKey

MEMBER-INTEREST Table:

mbr #	InterestName
*	*

BOOK Table:

BookID #	Title	InterestName
*		

BOOK COPY Table:

BookID #	CopyID #	CheckedOut?
*	*	

the library will have many copies of the same book.

AUTHORS Table:

AuthorID #	LastName	FirstName
*		

BOOK-AUTHORS Table:

BookID #	AuthorID #
*	*

CHECKOUTS Table:

CheckoutID #	mbr #	DateOut	DateDue
*			

BOOKS CHECKED OUT Table:

checkoutID #	BookID #
*	*

for each checkout event, there can be multiple books checked out

EMPLOYEES Table:

EmplID #	LastName	FirstName	Address	JobTitle	Salaried ?	Wage/week	Wage/hour	Bank	Routing #
*									

EMPLOYEES HOURS WORKED Table:

EmplID #	Date	Clock-In	Clock-out
----------	------	----------	-----------

Next step would be to normalize these tables to ensure compliance through 3NF.

Tables for DB in 3NF

Member Table:

mbr#	LastName	FirstName	Street Address	City	State	Zip	PhoneNum	DateJoined
*								

Interest Table:

Interest Name	Description
*	

Member-Interest Table:

mbr#	InterestName
*	*

Book Table:

BookID#	Title	InterestName
*		

Book-Copy Table:

BookID#	CopyID#	CheckedOut?
*	*	

Author Table:

AuthorID	LastName	FirstName
*		

Book-Author Table:

BookID#	AuthorID#
*	*

Checkout Table:

CheckoutID	mbr#	DateOut
*		

Books Checked Out Table:

checkoutID	BookID#
*	*

Employees Table:

EmplID	LastName	FirstName	Street Address	City	State	Zip	JobTitle	Salaried?	Bank	Routing#
*										

Salaried-Wage Table:

EmplID	Salary
*	

Hourly-Wage Table:

EmplID	hourlyWage
*	

Employees-Hours-Worked Table:

EmplID	Date	CheckIn	CheckOut
*	*		

Step 3

normalization
sols.

MEMBER Table:

mbr #	LastName	FirstName	Address	PhoneNum	DateJoined
*					

INTEREST Table:

InterestName	Description
*	

↳ 1NF: separate into

- street Address
- city
- state
- Zip

*PrimaryKey

MEMBER-INTEREST Table:

mbr #	InterestName
*	*

BOOK Table:

BookID #	Title	InterestName
*		

BOOK COPY Table:

BookID #	CopyID #	CheckedOut?
*	*	

the library will have many copies of the same book.

AUTHORS Table:

AuthorID #	LastName	FirstName
*		

BOOK-AUTHORS Table:

BookID #	AuthorID #
*	*

CHECKOUTS Table:

CheckoutID #	mbr #	DateOut	DateDue
*			

3NF: not independent of DateOut. You can calculate Due Date based on the Date Out & length of checkout rules.
REMOVE this attribute.

BOOKS CHECKED OUT Table:

checkoutID #	BookID #
*	*

for each checkout event, there can be multiple books checked out

EMPLOYEES Table:

EmplID #	LastName	FirstName	Address	JobTitle	Salaried ?	Wage/week	Wage/hour	Bank	Routing #
*									

3NF: not independent of Wage/hour.
REMOVE this attribute

EMPLOYEES HOURS WORKED Table:

EmplID #	Date	Clock-in	Clock-out
----------	------	----------	-----------

probably want to create 2 additional tables to account for the salary of salaried employees & the wage for hourly employees.

salaried-wage-table:

*EmplID #	Salary
-----------	--------

hourly-wage-table:

*EmplID #	hourly wage
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Next step would be to normalize these tables to ensure compliance through 3NF.